

# Subsystem Specification Simulation & test

Let the whole system be exercised, failed on purpose and rehearsed at home, first against a simulated boat and then with more and more real hardware in the loop, with the same software throughout.

|          |                                                                           |
|----------|---------------------------------------------------------------------------|
| Document | BOATY-SSS-SIM                                                             |
| Issue    | Issue G (baseline candidate, SDR)                                         |
| Date     | 30 September 2026                                                         |
| Status   | Baseline candidate for the SDR freeze                                     |
| Parents  | SRS Issue H, ADD Issue H, ICD Issue H, FMEA Issue G, KCL Issue D          |
| Content  | 3 allocated SRS requirements → 30 subsystem requirements (23 M, 7 S, 0 C) |

# Document control

| Issue | Date              | Change                                                                                                                                                                                                                                                             | By                             |
|-------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| A     | 28 September 2026 | First issue, for review.                                                                                                                                                                                                                                           | Claude (drafted)               |
| B     | 28 September 2026 | Hardware-in-the-loop rigs L1-L3 added (SIM-D13 to D24). Test catalogue rebuilt with IDs and extended with FMEA-derived scenarios (BOATY-FMEA-001).                                                                                                                 | Claude, owner request          |
| C     | 28 September 2026 | L1-10 (adult PIN) and B-07 (cell acceptance) added for FMEA FM-48 and A-20.                                                                                                                                                                                        | Claude, owner decision (CR-03) |
| D     | 29 September 2026 | Simulator built (slices 1-2): our own boat model through SITL's JSON interface (ADD DD-20), B1 router topology, real services on a simulated Pi Zero. SIM-D01/02/07 updated; SIM-D25 to D28 added; test results shown against the catalogue; SC-02 at 3 s (CR-05). | Claude, owner decisions        |
| E     | 29 September 2026 | Slice 3: Mission Control in the loop. SC-41 to SC-43 added; SIM-D29 and D30 added; SC-31 and SC-33 evidence sources named. SC-33 run against the live Claude API (39 cases).                                                                                       | Claude, owner request          |
| F     | 30 September 2026 | SDR WP2/WP3: SC-27 built; SC-11, SC-12, SC-13 and SC-20 built and redefined as run; SC-08 recorded as an SWE-005 deviation (verified on rig L2).                                                                                                                   | Claude, SDR                    |
| G     | 30 September 2026 | SDR WP5 (RID-08): issue, parents and applicable documents taken from the baseline register; issued as the SDR baseline candidate.                                                                                                                                  | Claude                         |

| Role                | Name                      | Signature / date |
|---------------------|---------------------------|------------------|
| Author              | Claude, for Project Boaty |                  |
| Reviewer / approver | Project owner             |                  |

Review guidance: section 5 is what will be built and tested. Check that each requirement is right, sufficient and testable. Section 4 shows every parent requirement is covered (the build script checks this).

## Contents

|                                                   |          |
|---------------------------------------------------|----------|
| <b>1. Scope</b>                                   | <b>4</b> |
| 1.1 Purpose . . . . .                             | 4        |
| 1.2 Boundary . . . . .                            | 4        |
| 1.3 Breakdown . . . . .                           | 4        |
| 2. Applicable documents . . . . .                 | 4        |
| <b>3. Interfaces</b>                              | <b>4</b> |
| <b>4. Parent requirements</b>                     | <b>5</b> |
| <b>5. Subsystem requirements</b>                  | <b>6</b> |
| 5.1 Simulator . . . . .                           | 6        |
| 5.2 Tests . . . . .                               | 6        |
| 5.3 CI and reproducibility . . . . .              | 6        |
| 5.4 Built in slices 1-2 (Issue D) . . . . .       | 6        |
| 5.5 Hardware-in-the-loop rigs (Issue B) . . . . . | 7        |
| <b>6. Test catalogue</b>                          | <b>8</b> |

|                                                                      |           |
|----------------------------------------------------------------------|-----------|
| 6.1 Failsafe scenarios (one per FS requirement) . . . . .            | 8         |
| 6.2 FMEA-derived simulator scenarios . . . . .                       | 8         |
| 6.3 Rig L1: real computers, simulated boat . . . . .                 | 10        |
| 6.4 Rig L2: iron bird . . . . .                                      | 10        |
| 6.5 Rig L3: water tank . . . . .                                     | 11        |
| 6.6 Bench, rehearsal and pool items referenced by the FMEA . . . . . | 11        |
| 6.7 What the rigs verify earlier . . . . .                           | 11        |
| <b>7. Design constraints and budgets</b>                             | <b>12</b> |
| <b>8. Verification and open items</b>                                | <b>12</b> |

# 1. Scope

## 1.1 Purpose

Let the whole system be exercised, failed on purpose and rehearsed at home, first against a simulated boat and then with more and more real hardware in the loop, with the same software throughout.

## 1.2 Boundary

| Inside this subsystem                                                                                                                                                                                                                                                                                                                                                                                                                                   | Outside (and who owns it)                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• ArduPilot Rover 4.7.1 SITL</li><li>• Boat and battery model (JSON physics)</li><li>• Radio-link relay and B1 router stand-in</li><li>• Simulated camera for B2</li><li>• Scenario runner and test suites (software/tests)</li><li>• CI configuration</li><li>• Rig L1: real computers + SITL</li><li>• Rig L2: iron bird (real helm and power train)</li><li>• Rig L3: water-tank thrust and weed rig</li></ul> | <ul style="list-style-type: none"><li>• The software under test (MCN, MCP services)</li><li>• Pool and lake trials (per-subsystem and system test procedures)</li></ul> |

## 1.3 Breakdown

| Item                     | Description                                                                                                                           |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| SIM-1 Simulator          | SITL + boat model (software/boaty/sim) + launcher; services run from software/boaty/mcp                                               |
| SIM-2 Test suites        | Unit, contract, scenario, FMEA-derived, LLM evaluation                                                                                |
| SIM-3 CI                 | GitHub Actions workflows                                                                                                              |
| SIM-4 Rig L1             | Real Pi 5 panel, real Pi Zero + camera, real Wi-Fi; SITL on a laptop, reaching the Pi Zero UART through a USB-serial adapter          |
| SIM-5 Rig L2 'iron bird' | Real FC, firmware and parameters, power board with key, ESCs and motors, GNSS, beacon, on a board; driven by the real Mission Control |
| SIM-6 Rig L3 tank        | One pod in a water tub on a load cell, with current logging and real pond weed                                                        |

## 2. Applicable documents

| Document               | Use                                                           |
|------------------------|---------------------------------------------------------------|
| BOATY-SRS-001 Issue H  | Parent requirements                                           |
| BOATY-ADD-001 Issue H  | Allocation, decisions, budgets, early verification items      |
| BOATY-ICD-001 Issue H  | Interface definitions (all listed in section 3 apply in full) |
| BOATY-FMEA-001 Issue G | Failure modes and actions carried into this issue             |
| BOATY-KCL-001 Issue D  | Component data, parameters and cost                           |

## 3. Interfaces

The subsystem shall meet every interface below as defined in the ICD. 'Owner' means this subsystem maintains the definition.

| Interface             | Other end       | Role     | ICD |
|-----------------------|-----------------|----------|-----|
| IF-15 Site data       | Mission Control | Conforms | \$5 |
| IF-21 Simulation link | Mission Control | Owner    | \$5 |

## 4. Parent requirements

**Primary (3):** this subsystem is responsible for meeting and verifying these SRS requirements.

| SRS            | Pri      | Requirement (abridged)                                                                                                    | Met by                                                                 |
|----------------|----------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>SWE-004</b> | <b>M</b> | The whole system shall run in simulation (helm simulator and simulated camera) on the Pi 5 or a laptop, with Mission...   | SIM-D01, SIM-D02, SIM-D03, SIM-D04, SIM-D25, SIM-D26, SIM-D29, SIM-D13 |
| <b>SWE-005</b> | <b>M</b> | Each failsafe requirement (FS-001 to FS-013) shall have an automated simulation scenario. Deviation: FS-008 (ESCs stop... | SIM-D06, SIM-D07, SIM-D27, SIM-D28, SIM-D30                            |
| <b>SWE-007</b> | <b>S</b> | Unit tests (pytest), linting (ruff) and type checking (mypy) shall run automatically on every push.                       | SIM-D11                                                                |

## 5. Subsystem requirements

Requirements are numbered SIM-Dnn. Pri, Ver and Stage follow SRS conventions (M/S/C; Test, Demonstration, Inspection, Analysis, Simulation; SIM, BENCH, POOL, LAKE). 'Trace' lists the parent SRS requirements, interfaces (IF), decisions (DD) and early verification items (V) it implements.

### 5.1 Simulator

| ID             | Requirement                                                                                                                                                                         | Pri | Ver | Stage | Trace                |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-------|----------------------|
| <b>SIM-D01</b> | SITL shall run the same ArduPilot version as the helm (Rover 4.7.1) and the controlled parameter file, flying our own skid-steer boat model through the JSON interface (ADD DD-20). | M   | D   | SIM   | SWE-004, V-12, DD-20 |
| <b>SIM-D02</b> | The real B2/B3 code shall serve IF-03 in simulation, with a simulated camera rendering images at the boat's true position.                                                          | M   | T   | SIM   | SWE-004, IF-21       |
| <b>SIM-D03</b> | One command shall start the simulator on a laptop or the Pi 5 at a given site's home.                                                                                               | M   | D   | SIM   | SWE-004              |
| <b>SIM-D04</b> | Mission Control shall run against the simulator with configuration changes only.                                                                                                    | M   | D   | SIM   | SWE-004, SWE-003     |
| <b>SIM-D05</b> | A rehearsal mode shall run the physical panel against the simulator for crew practice.                                                                                              | S   | D   | SIM   | OPS-012              |

### 5.2 Tests

| ID             | Requirement                                                                                                                                                                                                                                                        | Pri | Ver | Stage | Trace            |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-------|------------------|
| <b>SIM-D06</b> | Each FS-001 to FS-013 requirement shall have an automated scenario with explicit pass criteria (SC-01 to SC-13, section 6). FS-008 is proven on rig L2.                                                                                                            | M   | T   | SIM   | SWE-005          |
| <b>SIM-D07</b> | Fault injection shall use SITL simulation parameters (SIM_GPS1_ENABLE, SIM_MAG1_ORIENT), boat-model faults (motor loss or reversal, weed drag, wind, battery drain, key out) and radio-link impairment (cut, loss, latency), mapped per scenario in the test code. | M   | T   | SIM   | SWE-005, IF-21   |
| <b>SIM-D08</b> | Contract tests for IF-03 and IF-14 shall run against both the simulator and real hardware.                                                                                                                                                                         | M   | T   | SIM   | IF-14, IF-03     |
| <b>SIM-D09</b> | The validator's adversarial suite ( $\geq 50$ cases) and a property-based comparison against an independent geometry implementation shall run in CI (SC-31).                                                                                                       | M   | T   | SIM   | VAL-006          |
| <b>SIM-D10</b> | The LLM evaluation set ( $\geq 30$ instructions, including adversarial child phrasing) shall run on demand with a per-run cost cap (SC-33).                                                                                                                        | S   | T   | SIM   | NLI-003, NLI-006 |

### 5.3 CI and reproducibility

| ID             | Requirement                                                                                            | Pri | Ver | Stage | Trace   |
|----------------|--------------------------------------------------------------------------------------------------------|-----|-----|-------|---------|
| <b>SIM-D11</b> | Every push shall run ruff, mypy and unit tests. SITL scenarios shall run on pull requests and nightly. | S   | I   | SIM   | SWE-007 |
| <b>SIM-D12</b> | Tool and dependency versions (SITL commit, Python lock file) shall be pinned in the repository.        | M   | I   | SIM   | SWE-006 |

### 5.4 Built in slices 1-2 (Issue D)

| ID             | Requirement                                                                                                                                                                                                   | Pri | Ver | Stage | Trace          |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-------|----------------|
| <b>SIM-D25</b> | The boat model shall take its parameters from the Key Component List (mass, thrust ahead/astern, drag, windage, pack capacity and resistance) and be re-fitted to thrust-stand and pool data when they exist. | M   | A   | SIM   | SWE-004, DD-20 |

| ID             | Requirement                                                                                                                                                                                                                                                       | Pri | Ver | Stage | Trace                 |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-------|-----------------------|
| <b>SIM-D26</b> | The simulated boat shall have the boat's MAVLink topology: SITL's port joined by a B1 router to the radio relay and to the services, so a mission-computer failure takes the bank link with it.                                                                   | M   | D   | SIM   | SWE-004, IF-04, IF-21 |
| <b>SIM-D27</b> | Every timing criterion shall be measured in simulated time from truth (commanded outputs, thrust, position), not from telemetry round trips.                                                                                                                      | M   | I   | SIM   | SWE-005               |
| <b>SIM-D28</b> | Each run shall write per-scenario evidence (measured values against the criterion) and a report (software/results). Known native gaps are marked as such, never hidden.                                                                                           | M   | I   | SIM   | SWE-005, SAF-007      |
| <b>SIM-D29</b> | Mission Control shall run in the simulator as it runs on the Pi 5 (same code, simulated time), with Claude's replies from a stand-in client that produces the API's response format. The live API is exercised separately by the recorded evaluation set (SC-33). | M   | I   | SIM   | SWE-004, IF-11        |
| <b>SIM-D30</b> | Evidence that needs no simulator (validator coverage, property-based runs) shall be written to software/results/unit_evidence.json and reported with the SITL results.                                                                                            | S   | I   | SIM   | VAL-006, SWE-005      |

## 5.5 Hardware-in-the-loop rigs (Issue B)

| ID             | Requirement                                                                                                                                                                                                               | Pri | Ver | Stage | Trace                                            |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-------|--------------------------------------------------|
| <b>SIM-D13</b> | Rig L1 shall connect SITL to the real Pi Zero 2W through a 3.3 V USB-serial adapter at 115200 baud on the Pi's UART, so the production router configuration is used unchanged. The real Pi 5 panel and Wi-Fi are used.    | M   | D   | BENCH | SWE-004, IF-04, IF-01                            |
| <b>SIM-D14</b> | Rig L1 shall timestamp STOP from button edge to the SITL mode change with $\leq 10$ ms resolution, over $\geq 100$ trials (L1-03).                                                                                        | M   | T   | BENCH | FS-009, COM-003                                  |
| <b>SIM-D15</b> | Rig L1 shall switch the Pi Zero's 5 V supply under script control for power-cut endurance testing (L1-08).                                                                                                                | S   | T   | BENCH | IF-08                                            |
| <b>SIM-D16</b> | Rig L1 shall allow the sealed box with the Pi Zero to be heat-soaked at 30 °C (heat lamp or sun) while running a mission (L1-09).                                                                                         | S   | T   | BENCH | PWR-010                                          |
| <b>SIM-D17</b> | Rig L2 shall mount the real FC (flashed firmware and controlled parameters), power board with fuse and magnetic key, ESCs, motors, GNSS with sky view, and beacon, driven by the real Mission Control over the real link. | M   | D   | BENCH | SAF-001, IF-05, IF-06, IF-07, IF-09              |
| <b>SIM-D18</b> | Rig L2 shall support physical fault injection: GNSS unplug and data-line cut, AP off, bench-supply voltage ramp, ESC signal cut, key removal, FC power-cycle, and thrust steps.                                           | M   | T   | BENCH | FS-001, FS-002, FS-004, FS-008, MOD-003, PWR-005 |
| <b>SIM-D19</b> | Rig L2 shall record motor current, 5 V rails and ESC signals (logic analyser or scope), and detect motor stop within 50 ms (ESC telemetry or optical).                                                                    | M   | T   | BENCH | FS-008, FS-009, PWR-005                          |
| <b>SIM-D20</b> | Rig L2 shall run with props removed in air, or with pods submerged in a covered tank. Removing the key is the emergency stop, and the adult-only rules apply.                                                             | M   | I   | BENCH | OPS-007, CHD-003                                 |
| <b>SIM-D21</b> | Rig L3 shall measure one pod's thrust ( $\pm 0.05$ N) and current, forward and astern, at 0-100% in 10% steps (L3-01).                                                                                                    | M   | T   | BENCH | NAV-001, ENV-002, IF-06                          |
| <b>SIM-D22</b> | Rig L3 shall introduce real pond weed at the inlet and record whether the weed-shedding routine clears it (L3-02).                                                                                                        | S   | T   | BENCH | ENV-005, FS-006                                  |
| <b>SIM-D23</b> | Every rig test shall be recorded with date, software and firmware versions, parameter hash and results, and committed to the repository.                                                                                  | M   | I   | BENCH | SWE-006                                          |
| <b>SIM-D24</b> | Every FMEA failure mode with severity $\geq 8$ shall be exercised by at least one SIM, rig or bench test in the catalogue. The build checks this.                                                                         | M   | A   | SIM   | SAF-006, FS-013                                  |

## 6. Test catalogue

Every test has an ID, which the FMEA (BOATY-FMEA-001) and the test procedures reference. SC = simulator scenario; L1/L2/L3 = rig tests; B = bench; R = rehearsal; P = pool. Injection parameter names are fixed per SITL version and kept in the scenario code. **Result** is read from software/results/sitl\_results.json (the last full SITL run); '-' means not yet built.

### 6.1 Failsafe scenarios (one per FS requirement)

| ID    | Test                                     | Injection / method                                                                                                    | Pass criterion                                                      | Refs                 | Result                                    |
|-------|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|----------------------|-------------------------------------------|
| SC-01 | Battery drain to 35% then 15%            | Simulated battery drain                                                                                               | RTL at 35%; alarm at 15%; reaches home                              | FS-001               | Pass                                      |
| SC-02 | Link cut in STEERING                     | Drop UDP 14550                                                                                                        | HOLD $\leq 3$ s (CR-05); RTL at 10 s (B4)                           | FS-002               | Pass                                      |
| SC-03 | Link cut in AUTO                         | Drop UDP 14550                                                                                                        | Mission continues; RTL at 60 s (B4)                                 | FS-003, FM-31        | Pass                                      |
| SC-04 | GNSS failure 5 s then restore            | SITL GPS failure                                                                                                      | Motors stop $\leq 3$ s; RTL after 10 s healthy (B6)                 | FS-004, FM-01, FM-06 | Pass                                      |
| SC-05 | Heavy drag at throttle $\geq 50\%$       | SITL drag / wind                                                                                                      | HOLD $\leq 5$ s plus event                                          | FS-005, FM-16        | Known gap (via V-05; B7 backs up at 10 s) |
| SC-06 | Drag released after burst 2              | SITL + B5                                                                                                             | $\leq 3$ bursts; resumes mission                                    | FS-006, FM-27        | Pass                                      |
| SC-07 | Kill all MCP services mid-mission        | Stop stub and B-services                                                                                              | Mission completes; RTL; HOLD at home                                | FS-007, FM-26        | Pass                                      |
| SC-08 | Helm signal loss to ESCs                 | Not simulated: the behaviour is in the ESC firmware. Verified on rig L2 (L2-10); SWE-005 deviation                    | See L2-10                                                           | FS-008               | -                                         |
| SC-09 | STOP in every state                      | Panel emulation                                                                                                       | Motors stop $\leq 1$ s                                              | FS-009               | Pass (panel emulation, 4 armed states)    |
| SC-10 | Moisture flagged                         | Stub health = moisture                                                                                                | RTL $\leq 2$ s                                                      | FS-010, FM-24        | Pass                                      |
| SC-11 | GNSS loss and battery failsafe together  | GNSS off 25 s during a battery RTL, and before the battery crosses 35%                                                | Motors off $\leq 3$ s and stay off until the fix is back; then home | FS-011               | Pass                                      |
| SC-12 | Log and announce audit                   | Mission Control in the loop; battery, GNSS, water and weed failsafes                                                  | Each logged and announced $\leq 2$ s after the boat acts            | FS-012               | Pass                                      |
| SC-13 | Single-fault sweep across mission phases | 7 faults (GNSS loss, 20 m GNSS offset, compass, motor, link, mission computer, weed) $\times 3$ phases near the fence | Never outside the fence under power                                 | FS-013               | Pass                                      |

### 6.2 FMEA-derived simulator scenarios

| ID    | Test                                     | Injection / method                                       | Pass criterion                               | Refs  | Result |
|-------|------------------------------------------|----------------------------------------------------------|----------------------------------------------|-------|--------|
| SC-20 | GNSS glitch: 20-50 m jump near the fence | SITL GPS glitch, in and out, 2 s; 20 m for 10 s and 30 s | No uncommanded exit; glitch rejected or HOLD | FM-02 | Pass   |

| ID    | Test                                          | Injection / method                                                                          | Pass criterion                                             | Refs                  | Result                                                  |
|-------|-----------------------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------|-----------------------|---------------------------------------------------------|
| SC-21 | GNSS frozen: stale position while moving      | SITL GPS freeze                                                                             | Detected; motors stop $\leq 3$ s                           | FM-03                 | -                                                       |
| SC-22 | Compass offset 30° and 90°                    | SITL compass offset                                                                         | Yaw fallback or HOLD; stays inside the fence               | FM-04, FM-05          | -                                                       |
| SC-23 | Helm restart mid-mission                      | Restart SITL process                                                                        | Motors stop; boots disarmed; alarm at Mission Control      | FM-07                 | -                                                       |
| SC-24 | Parameter differs from baseline               | Change one failsafe parameter                                                               | Arming blocked; difference shown                           | FM-09                 | Pass                                                    |
| SC-25 | Fence missing or disabled                     | Skip the fence upload                                                                       | Arming refused                                             | FM-10                 | Pass                                                    |
| SC-26 | Site file lat/lon swapped; wrong site         | Corrupt the site file                                                                       | Linter and pre-arm both refuse                             | FM-11                 | Pass                                                    |
| SC-27 | RTL from the far side of the island           | RTL where the straight line home crosses the island; calm and 3 m/s wind pushing towards it | Never enters a zone or leaves the fence; home $\leq 180$ s | FM-13, FM-59, NAV-007 | Pass                                                    |
| SC-28 | Persistent breach (wind pushing out)          | Strong offshore wind                                                                        | Motors stop by 30 s / 10 m                                 | FM-14                 | Pass                                                    |
| SC-29 | Single motor failure                          | Zero one output                                                                             | Divergence watchdog → HOLD $\leq 20$ s                     | FM-17                 | Pass                                                    |
| SC-30 | Boat-service command fuzz                     | Random forbidden commands                                                                   | All blocked by the filter                                  | FM-27                 | Pass (unit fuzz, 6,000 cases)                           |
| SC-31 | Validator adversarial + property-based        | Hypothesis vs independent geometry                                                          | No accepted mission crosses a boundary                     | FM-34                 | Pass (400 generated, 118 accepted, closest 3.273 m)     |
| SC-32 | Read-back corruption                          | Alter one item in transfer                                                                  | GO stays disabled                                          | FM-35, FM-47          | Pass                                                    |
| SC-33 | LLM evaluation with adversarial phrasing      | Recorded instruction set                                                                    | Schema-valid; must-decline declined                        | FM-36, FM-37          | Pass (live API: 41/41 cases, declines 14/14, p95 6.9 s) |
| SC-34 | Link degradation: 30/60/90% loss, 2 s latency | Network impairment                                                                          | FS-002/003 behaviour holds; STOP latency logged            | FM-39, FM-40          | -                                                       |
| SC-35 | Home sanity                                   | Arm with GNSS unsettled / far from site home                                                | Refused or warned before GO                                | FM-46                 | Pass                                                    |
| SC-36 | Interrupted upload                            | Drop the link mid-transfer                                                                  | No partial mission accepted                                | FM-47                 | -                                                       |
| SC-37 | Foreign GCS heartbeat                         | Second system-255 source                                                                    | C7 refuses to operate; alarm                               | FM-41                 | Pass                                                    |
| SC-38 | First-motion heading check                    | Compass reversed                                                                            | HOLD $\leq 10$ s after start                               | FM-05, FM-18          | Pass                                                    |
| SC-39 | Capacity overstated by 30%                    | Wrong capacity parameter                                                                    | Voltage backstop triggers RTL in time                      | FM-15                 | -                                                       |
| SC-40 | Arm lands in HOLD                             | Arm with default RC input present                                                           | Armed in HOLD, motors off                                  | FM-53                 | Pass                                                    |

| ID    | Test                             | Injection / method                                             | Pass criterion                                                                                                   | Refs                      | Result |
|-------|----------------------------------|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|---------------------------|--------|
| SC-41 | Instruction to captain's log     | Typed instruction, stand-in Claude reply, full Mission Control | Validated, read back equal, flown inside the fence, home, auto-disarm at 60 s, photos sha256-verified before ack | SWE-004, VAL-010, MCN-D47 | Pass   |
| SC-42 | Persistent breach with B7 absent | Gale; boat services not running                                | C1 stops motors before 30 s or 10 m outside (trigger 8 m)                                                        | FM-14, MCN-D59            | Pass   |
| SC-43 | Boat stops itself                | Dead motor mid-mission                                         | Reason spoken $\leq 2$ s after HOLD; resume needs the PIN                                                        | MCN-D60                   | Pass   |

### 6.3 Rig L1: real computers, simulated boat

| ID    | Test                                            | Injection / method                                         | Pass criterion                                                                   | Refs          | Result |
|-------|-------------------------------------------------|------------------------------------------------------------|----------------------------------------------------------------------------------|---------------|--------|
| L1-01 | End-to-end mission, real computers              | Voice $\rightarrow$ captain's log over real Wi-Fi and UART | Completes; all artefacts logged                                                  | SWE-004       | -      |
| L1-02 | Panel self-test and state $\times$ button table | Real buttons                                               | Matches ICD IF-12                                                                | FM-32, FM-33  | -      |
| L1-03 | STOP latency, 100 trials                        | Button edge $\rightarrow$ SITL mode                        | $P99 \leq 0.5$ s                                                                 | FM-40, FS-009 | -      |
| L1-04 | Photo sync, 200 photos                          | Real Wi-Fi at 10 m                                         | $\leq 5$ min; hashes verified                                                    | CAM-007       | -      |
| L1-05 | UART load                                       | Maximum telemetry + services                               | GCS heartbeat jitter $< 0.5$ s                                                   | FM-28         | -      |
| L1-06 | Mission Control power pulled mid-mission        | Unplug the Pi 5                                            | B4 RTL at 60 s                                                                   | FM-31         | -      |
| L1-07 | Range walk on land                              | Walk the Pi Zero away                                      | RSSI vs distance; flap behaviour as SC-34                                        | FM-39         | -      |
| L1-08 | Pi Zero power-cut $\times 50$                   | Switched 5 V                                               | No filesystem damage                                                             | FM-26         | -      |
| L1-09 | Sealed-box heat soak 30 $^{\circ}\text{C}$      | Heat lamp                                                  | No throttling for 60 min                                                         | PWR-010       | -      |
| L1-10 | Adult PIN: unlock, timeout, lockout             | Web UI                                                     | 10 min unlock; 5 failures $\rightarrow$ 5 min lockout; motors still need the key | FM-48, MC-008 | -      |

### 6.4 Rig L2: iron bird

| ID    | Test                               | Injection / method                 | Pass criterion                       | Refs           | Result |
|-------|------------------------------------|------------------------------------|--------------------------------------|----------------|--------|
| L2-01 | Motor direction and channel map    | Props off                          | Left/right and sense correct         | FM-18          | -      |
| L2-02 | Arming gated on the key            | Key in/out                         | Refused with key out; rail $< 0.5$ V | FM-22, MOD-003 | -      |
| L2-03 | GNSS unplugged                     | Pull connector                     | EKF failsafe HOLD                    | FM-01          | -      |
| L2-04 | GNSS data line cut, power on       | Cut TX                             | Detected $\leq 3$ s                  | FM-03          | -      |
| L2-05 | Compass vs motor current           | Motor-interference calibration     | Interference $\leq 30\%$             | FM-04          | -      |
| L2-06 | Heading vs reference               | Known bearing                      | Within $10^{\circ}$                  | FM-05          | -      |
| L2-07 | FC power-cycle with motors running | Interrupt FC supply                | Motors stop; boots disarmed          | FM-07          | -      |
| L2-08 | Thrust-step brownout               | Full reverse $\rightarrow$ forward | 5 V rails in regulation              | FM-08, PWR-005 | -      |

| ID    | Test                       | Injection / method  | Pass criterion                             | Refs                 | Result |
|-------|----------------------------|---------------------|--------------------------------------------|----------------------|--------|
| L2-09 | Supply voltage ramp        | Bench supply        | Battery failsafe at thresholds             | FM-15                | -      |
| L2-10 | ESC signal cut and restore | Cut the signal lead | Stop $\leq 1$ s; restart only from neutral | FM-19, FM-20, FS-008 | -      |
| L2-11 | STOP to motors stopped     | Panel STOP          | $\leq 1$ s                                 | FS-009, FM-40        | -      |
| L2-12 | Beacon patterns            | Cycle states        | Each state distinguishable                 | REC-004              | -      |
| L2-13 | Key-switch short detection | Bridge the MOSFET   | Checklist rail test flags it               | FM-22                | -      |

## 6.5 Rig L3: water tank

| ID    | Test                      | Injection / method   | Pass criterion                                 | Refs           | Result |
|-------|---------------------------|----------------------|------------------------------------------------|----------------|--------|
| L3-01 | Thrust and current curves | Load cell, 0-100%    | $\geq 2$ N forward per pod; current $\leq 8$ A | NAV-001, IF-06 | -      |
| L3-02 | Weed fouling trials       | Real pond weed       | Cleared in $\geq 2$ of 3                       | FM-16, ENV-005 | -      |
| L3-03 | Cruise power estimate     | Thrust vs drag model | $\leq 10$ W at 1.0 m/s                         | PWR-009        | -      |

## 6.6 Bench, rehearsal and pool items referenced by the FMEA

| ID   | Kind      | Test                           | Pass criterion                             | Refs                       |
|------|-----------|--------------------------------|--------------------------------------------|----------------------------|
| B-01 | BENCH     | Prop guard probe               | No blade contact                           | FM-21, FM-45               |
| B-02 | BENCH     | BMS protection                 | BMS trips; fuse intact                     | FM-23                      |
| B-03 | BENCH     | Box dunk                       | No ingress                                 | FM-24                      |
| B-04 | BENCH     | Harness pull and connector     | No disconnect                              | FM-25                      |
| B-05 | BENCH     | Camera failure                 | Health reports fault; mission unaffected   | FM-29                      |
| B-07 | BENCH     | Salvaged cell acceptance       | Only compliant cells accepted and recorded | FM-23                      |
| B-06 | BENCH     | Clock without GNSS             | Timestamps flagged until GNSS time         | FM-30                      |
| R-01 | REHEARSAL | Operator contingency rehearsal | All section 3.5 scenarios rehearsed        | FM-42, FM-43, FM-44, FM-12 |
| P-01 | POOL      | Internet absent                | Templates offered                          | FM-38                      |

## 6.7 What the rigs verify earlier

| Item                                  | First verified (Issue A) | Now also by                                | Still needs                  |
|---------------------------------------|--------------------------|--------------------------------------------|------------------------------|
| FS-009 STOP $\leq 1$ s                | BENCH / POOL             | L1-03 (to helm), L2-11 (to motors stopped) | On-water confirmation (POOL) |
| COM-003 telemetry and latency         | POOL                     | L1-03, L1-05                               | Range effects (LAKE)         |
| CAM-007 / MCP-D16 photo sync          | POOL                     | L1-04                                      | -                            |
| MCN-D21 link quality display          | POOL                     | L1-07                                      | -                            |
| MCN-D22 manual drive                  | POOL                     | L2 (motors follow the joystick)            | Handling (POOL)              |
| NAV-001 bidirectional thrust          | POOL                     | L2-01, L3-01                               | -                            |
| PRP-D02 thrust $\geq 4$ N             | BENCH (estimate)         | L3-01 (measured)                           | -                            |
| PRP-D03/D04 cruise throttle and power | POOL                     | L3-03 (estimate from data)                 | Confirm (POOL)               |
| PRP-D11 / ENV-005 weed                | LAKE                     | L3-02 (real weed)                          | Confirm (LAKE)               |
| FS-005/006 stuck and shedding         | POOL                     | SC-05/06 + L3-02                           | Confirm (POOL)               |
| FS-001 / FS-004 on real hardware      | SIM only                 | L2-09, L2-03/04                            | -                            |
| REC-D03 beacon patterns               | LAKE                     | L2-12 (patterns)                           | Visibility (LAKE)            |

| Item            | First verified (Issue A) | Now also by                 | Still needs |
|-----------------|--------------------------|-----------------------------|-------------|
| PWR-010 thermal | BENCH                    | L1-09 under a real workload | -           |

The FMEA check (SIM-D24) confirms that every failure mode with severity  $\geq 8$  has at least one test here.

## 7. Design constraints and budgets

- Same parameter file as the real helm, plus sitl.parm
- All timings in simulated time; runs faster than real time (5x used)
- Runs on a laptop or the Pi 5
- Rig L2 is run with props off, or in water with the tank covered. Removing the magnetic key is the emergency stop.

| Budget         | Allocation                                                                                                                                         |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Cost           | £0 software (open source). Rig equipment, outside the Mk1 BOM: USB-serial adapter £3, load cell + HX711 £6, tub; bench power supply assumed owned. |
| LLM evaluation | API credit, capped per run                                                                                                                         |
| Mass / power   | n/a                                                                                                                                                |

## 8. Verification and open items

Each requirement is verified at its stage using the method shown. Test procedures are written per stage and reference these IDs.

| Stage | Test | Demo | Inspect | Analysis | Sim | Reqs      |
|-------|------|------|---------|----------|-----|-----------|
| SIM   | 6    | 5    | 6       | 2        | -   | <b>19</b> |
| BENCH | 7    | 2    | 2       | -        | -   | <b>11</b> |
| POOL  | -    | -    | -       | -        | -   | <b>0</b>  |
| LAKE  | -    | -    | -       | -        | -   | <b>0</b>  |

| Ref      | Open item                                                                                                      |
|----------|----------------------------------------------------------------------------------------------------------------|
| Model    | Boat-model thrust, drag and inertia are estimates until the thrust stand and pool replace them (KCL section 8) |
| Slice 3  | Mission Control against the simulator; C7 impostor-GCS detection (SC-37)                                       |
| SIM-OI-1 | Rig L2 needs the first parts order (FC, ESCs, motors, power parts). Rig L1 needs only a Pi Zero 2W and camera. |